

SHEIKH HEFZUL BARI

Email: shbari.bd@gmail.com

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Education

PhD Symbiotic Environment Systems, Fukushima University, Japan (Expected to graduate in 2026)

Thesis title: ‘Monitoring and data analysis of suspended sediment transport from the Abukuma River watershed to the Pacific Ocean’

Supervisor: Dr. Yoshiyuki Yokoo

Synopsis: A combination of literature review, experimental research, and statistical analysis of suspended sediment dynamics in a torrential watershed.

MSc. Civil & Environmental Engineering, Shahjalal Uni. of Science & Technology (2014)

BSc. Civil & Environmental Engineering, Shahjalal Uni. of Science & Technology (2011)

Scholarships and Awards

Japanese Government (MEXT) Scholarship – PhD Fellowship

National Science and Technology (NST) Fellowship – MSc., Ministry of Science and Technology, Bangladesh

Best Researcher Award 2017– Leading University, Bangladesh

Merit-Based Scholarships – Ministry of Education, Bangladesh (Secondary & Higher Secondary Scholarships)

Technical Skills

Programming **Python** user for scientific plotting, data processing, and analysis of small to medium size datasets.

Languages Native in **Bengali**, communicative in **English** with the capacity for scientific writing, and basic proficiency in **Japanese** and **Hindi** for practical communication.

Research Grants

Estimating probable maximum precipitation for Bangladesh (As PI)

Leading University Research Grant (Partially Funded). Time: January 2019 - December 2019

Assessing climate change impact in Haor Basin of Bangladesh (As PI)

Funded by Leading University Research Grant. Time: January 2017 - December 2017

Development of low-cost desalination device (As PI)

Ministry of Science and Technology, Bangladesh. Time: January 2015 - March 2016

Research and Academic Experiences

Research Assistant Watershed Hydrology lab, Fukushima University, April, 2023 – March 2026

Faculty Member (On study Leave) Department of Civil Engineering, Leading University, Sylhet, Bangladesh, August 2013 – Present

Conference Presentations and Lectures / Invited Talk

****Bari, S.H. and Yokoo, Y., (2025) *A reconnaissance study on estuarine suspended sediment dynamics in the Abukuma River, Japan*. Invited oral presentation at the Japan Geoscience Union (JpGU) Meeting 2025, Chiba, Japan, 26–31 May.**

Publications (Journal:14, Conference: 8, Chapter:2, Only selected are listed here)

Journals

Bari, SH., Yokoo, Y. and Leong, C. (Accepted). Temporal dynamics and threshold behaviors of suspended sediment transport in the Abukuma River, Japan. *Hydrological Research Letters*.

Ugajin R, Yokoo Y, Sayama T, **Bari, SH.** and Leong C (Accepted). Improving Rainfall-Runoff-Inundation Model for realistic representations of main river with levees. *Hydrological Research Letters*.

Bari, SH., Yokoo, Y., Tateno, M. and Leong, C. (2025). Dynamics, particle size distribution and mineral composition of suspended sediment transport in the Abukuma River estuary. *Hydrological Research Letters*, 19(1), pp.51-57.

Bari, SH., Yokoo, Y. and Leong, C. (2024). A brief review of recent global trends in suspended sediment estimation studies. *Hydrological Research Letters*, 18(2), pp.51-57.

Husna, NEA., **Bari, SH.**, Islam, GT. and Islam, AS. (2024). Groundwater drought assessment in Southwestern Bangladesh. *Int. J. of Hydrology Science and Technology*, 17(3), pp.267-277.

Husna, NEA., **Bari, SH.**, Islam, GT., Islam, AS. and Hussain, MM. (2024). Modelling drought as a climate change indicator in the southwest coastal region of Bangladesh. *Int. J. of Hydrology Science and Technology*, 18(4), pp.351-67.

Hussain, MM., Mahmud, I. and **Bari, SH.** (2023). pyHomogeneity: a python package for homogeneity test of time series data. *J. of Open Research Software*, 11(1).

Bari, SH. and Husna, NEA. (2023). People's perception on climate change effects and adaptation in the Haor basin of Bangladesh. *Int. J. Hydrology Science and Technology*, 15(1), pp.40-53.

Bari SH. (2021). Estimating probable maximum precipitation for Bangladesh. *Int. J. Hydrology Science and Technology*, 11(4), pp.415-421.

Bari, SH., Hussain, MM. and Husna, NEA. (2017). Rainfall variability and seasonality in northern Bangladesh. *Theoretical and Applied Climatology*, 129(3-4), pp.995-1001.

Mahmud, I., **Bari, SH.** and Rahman, MT. (2017). Monthly rainfall forecast of Bangladesh using autoregressive integrated moving average method. *Environmental Engineering Research*, 22(2), pp.162-168.

Bari, SH., Hoque, MA., Rahman, MT. and Hussain, MM. (2016). Analysis of seasonal and annual rainfall trends in the northern region of Bangladesh. *Atmospheric Research*, 176-177, pp.148-158.

Husna, NEA., **Bari, SH.**, Hussain, MM., Rahman, MT. and Rahman, M. (2016). Ground water level prediction using artificial neural network. *Int. J. Hydrology Science and Technology*, 6(4), pp.371-381.

Hussain, MM., **Bari, SH.**, Tarif, ME., Rahman, MT. and Hoque, MA. (2016). Temporal and spatial variation of groundwater level in Mymensingh district, Bangladesh. *Int. J. Hydrology Science and Technology*, 6(2), 188-197.

Bari, SH., Rahman, MT., Hussain, MM. and Ray, S. (2015). Forecasting Monthly Precipitation in Sylhet City Using ARIMA Model. *Civil and Environmental Research*, 7(1), 69-77.

Conferences

Bari, S.H. and Yokoo, Y., (2025) *Characterizing suspended sediment in the Abukuma River estuary: Composition, transport, and estimation*. Oral presentation at the Japan Society of Hydrology and Water Resources 2025 Research Presentation Conference, Akita, Japan, 17–19 September.

Bari, S.H. and Yokoo, Y., (2025) *Suspended sediment in the Abukuma River estuary, Japan: Composition, particle size, and hysteresis*. Oral presentation at the Asia Oceania Geosciences Society (AOGS) Annual Meeting 2025, Marina Bay Sands, Singapore, 27 July – 1 August.

Bari, S.H., Tateno, M. and Yokoo, Y. 2024. Suspended sediment transport in the Abukuma River estuary. *The 12th International Symposium on Water Environment Systems — with Perspective of Global Safety* (Poster presentation), Sendai, Japan, 21–22 November.

Bari, S.H., Tateno, M. and Yokoo, Y. 2024. Estimating sand, silt and clay transport at the Abukuma River estuary for a comprehensive understanding of suspended sediment dynamics from the watershed to the Pacific Ocean. *Japan Geoscience Union (JpGU) Meeting 2024* (Poster presentation), Chiba, Japan, 26–31 May.

Bari, S.H. and Yokoo, Y., (2025) *Preliminary field study for understanding estuarine suspended sediment dynamics at the Abukuma River, Japan*. Presented at the Japan Society of Civil Engineers Annual Meeting 2024, Sendai, Japan, 28 February – 1 March.